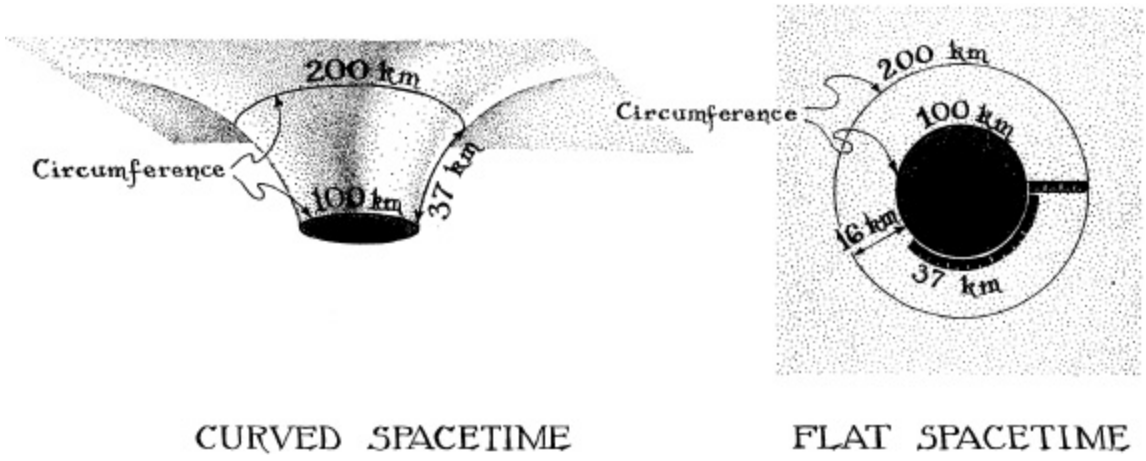
If space is actually flat around the black hole, but our perfect rulers are rubbery and thereby fool us into thinking space is curved, then the true geometry of space must be as shown on the right in Figure 11.1, and the true distance between the horizon and the circle must be 16 kilometers, as demanded by the flat-geometry laws of Euclid. However, general relativity insists that our perfect rulers not measure this true distance. Take a ruler and lay it down circumferentially around the hole just outside the horizon (curved thick black strip with ruler markings in right part of Figure 11.1). When oriented circumferentially like this, it does measure correctly the true distance. Cut the ruler off at 37 kilometers length, as shown. It now encompasses 37 percent of the distance around the hole. Then turn the ruler so it is oriented radially (straight thick black strip with ruler markings in Figure 11.1). As it is turned, general relativity requires that it shrink. When pointed radially, its true length must have shrunk to 16 kilometers, so it will reach precisely from the horizon to the outer circle. However, the scale on its shrunken surface must claim that its length is still 37 kilometers, and therefore that the distance between horizon and circle is 37 kilometers. People like Einstein who are unaware of the ruler's rubbery nature, and thus believe its inaccurate measurement, conclude that space is curved. However, people like you and me, who understand the rubberiness, know that the ruler has shrunk and that space is really flat.



11.1 Length measurements in the vicinity of a black hole from two different viewpoints. *Left:* Spacetime is regarded as truly curved, and perfect rulers measure precisely the lengths of the true spacetime. *Right:* Spacetime is regarded as truly flat and perfect rulers are rubbery. A 57-km-Iong perfect ruler, when oriented in a circumferential direction, measures precisely the lengths of the true, flat spacetime. However, when oriented radially, it shrinks by an amount that is greater the nearer it is to the hole, and therefore it reports radial lengths that are larger than the true ones (it reports 37 km rather than the true 16 km in the case shown).